Panchip Microelectronics Co., Ltd.

PAN3029 Hardware Design Reference

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Documentation

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Revision History

Version	Date	Description
V1.0	2023.06	Initial version
V1.1	2023.08	Update reference schematic design
V1.2	2023.10	1. Update schematics and BOM, 2. Add special notes
V1.3	2023.12	1. Update matching schematic diagram. 2. Add special points of attention
V1.4	2024.03	1. Update matching schematic

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1 Schematic Design Requirements

1.1 Power supply design requirements

The typical supply voltage of PAN3029 is 3.3V. The chip has two working modes. You can select LDO mode and DC-DC mode by configuring the working mode of the chip. The DC-DC mode requires a supply voltage range of 2-3.6V, and the LDO mode requires a supply voltage range of 1.8-3.6V. The following table lists the connection methods of the chip power-related pins and the recommended external device values.

The voltage range of writing eFuse inside the chip is 2.25V to 2.75V, and the typical value is 2.5V. Therefore, when writing eFuse, it is recommended to use an external VDD1 power supply of 2.5V. When reading eFuse, the power supply voltage supports 1.83-3.6V.

Table 1-1 Power-related pins Connectionand recommended external device values

PIN number	Name	Function	Signals		Value
			DC-DC Mode	Non-DC-DC Mode	
2	DCDC_V_FB	DCDC 输出	-	-	2.2uH+1uF+0.1uF
6	DVDD	数字电源 LDO 输出	-	-	1uF
7	VDD123	模拟电源	DCDC_V_FB	VCC	0.1uF
8	VBAT	模拟电源	VCC	VCC	10uF+0.1uF
9	VDDPA	PALDO 输出	-	-	0.1uF+100pF
27	VBAT_IO	数字电源	VCC	VCC	0.1uF
28	VBAT_DCDC	DCDC 电源	VCC	VCC	0.1uF

Note 1: In DCDC mode, the inductor and capacitor values of the pin DCDC_V_FB should be designed strictly according to the required 2.2uH+10uF+0.1uF. If the inductor and capacitor values are too small, the power consumption of the chip will increase.

1.2 Crystal design requirements

General requirements for external passive crystals are as follows:

1. Crystal frequency 32MHz;

2. ESR less than 50ohm;

3. The crystal load capacitance is recommended to be 12pF. If the load capacitance changes, the capacitance of the crystal in parallel to the ground in the schematic diagram needs to be changed;

4. Crystal frequency error ±10ppm;

5. Temperature (-20~70°C) frequency drift ±10ppm;

6. Excitation power 100uW.

It is recommended to use KDS passive crystal 1C232000AA0B. The specific parameters are shown below.

SPECIFICATION OF SURFACE MOUNT TYPE

1	.TYPE	DSX321G
2	.NOMINAL FREQUENCY	32.000000 MHz
3	.LOAD CAPACITANCE(CL)	12.0 pF
4	.DRIVE LEVEL	10 uW +/- 2 uW
5	.FREQUENCY TOLERANCE	+/- 10 × 10^{-6} at 25 deg.C +/- 3 deg.C
6	.SERIES RESISTANCE	50 ohms max. / CL = SERIES
7	.FREQUENCY CHARACTERISTICS OVER TEMPERATURE	+/- 10 × 10^{-6} / -20 deg.C to +70 deg.C (ref. +25 deg.C)
8	.OPERATING TEMPERATURE RANGE	-20 deg.C to +70 deg.C
9	.STORAGE TEMPERATURE RANGE	-40 deg.C to +85 deg.C
10	.SHUNT CAPACITANCE(C0)	2.0 pF max.
11	.INSULATION RESISTANCE	500 Mohms min. at 100V DC
12	.OVERTONE ORDER	Fundamental
13	.DIMENSIONS	Refer to Fig-1

The output of the active crystal oscillator is directly connected to the chip XC2 pin. The power supply of the active crystal oscillator can be provided by the chip GPIO or an external power supply. The operating voltage range supported by the active crystal oscillator is required to be 1.8~3.6V. For applications with a temperature range of -40~85°C, it is recommended to use KDS's active temperature compensation crystal 1XXD32000MBA. The specific parameters are shown in the following table.

(T_A=-40~+85°C, L_{OAD}_R//C=10kΩ//10pF, V_{CC}=+3.3V, unless otherwise noted)

	Item	Conditions	Limits			unit	Notes
			min.	typ.	max.		
1	Current Consumption		-	-	+2.0	mA	
2	Output Level		0.8	-	-	V_{P-P}	1
3	Symmetry	GND level (DC cut)	40/60	-	60/40	%	
4	Harmonics		-	-	-5	dBc	
5	Frequency Stability 1.Tolerance	After 2 times reflow Ref. to nominal frequency	-	-	±1.5	ppm	2,3
	2.vs Temperature	T_A=-30~+85°C Ref. to frequency (T_A=+25°C)	-	-	±0.5	ppm	
		T_A=-40~-30°C Ref. to frequency (T_A=+25°C)	-	-	±1.0	ppm	
	3.vs Supply Voltage	V_{CC}=+3.3V±5%	-	-	±0.2	ppm	
	4.vs Load Variation	L_{OAD}_R//C=(10kΩ//10pF)±10%	-	-	±0.2	ppm	
	5.vs Aging	T_A=Room ambient	-	-	±1.0	ppm/year	
6	Start Up Time	@90% of final Vout level	-	-	2.0	ms	
7	SSB Phase Noise	Relative to F0 level offset 1kHz	-	-	-125	dBc/Hz	

Under high transmission power, the chip will generate a lot of heat, which will be transferred to the crystal through the PCB. Since the frequency of the crystal will change with the temperature, it will cause frequency drift during the transmission process. This drift will pose a great challenge to the receiving demodulation. In severe cases, it will lead to receiving demodulation data errors and packet loss. Different BW and different SF have different requirements for crystals. Please refer to the "PAN3029 Crystal Selection Instructions" for details.

1.3 DC/DC design requirements

The inductor selection requires that the on-resistance is less than 100 m and the peak current is greater than 150mA (reference model: PIM252010-2R2MTS00).

1.4 Digital interface design requirements

1. SPI uses 4-wire CSN, SCK, MOSI and MISO interfaces. The maximum speed should be lower than 10Mbps. When operating efuse, it cannot exceed 8Mbps.

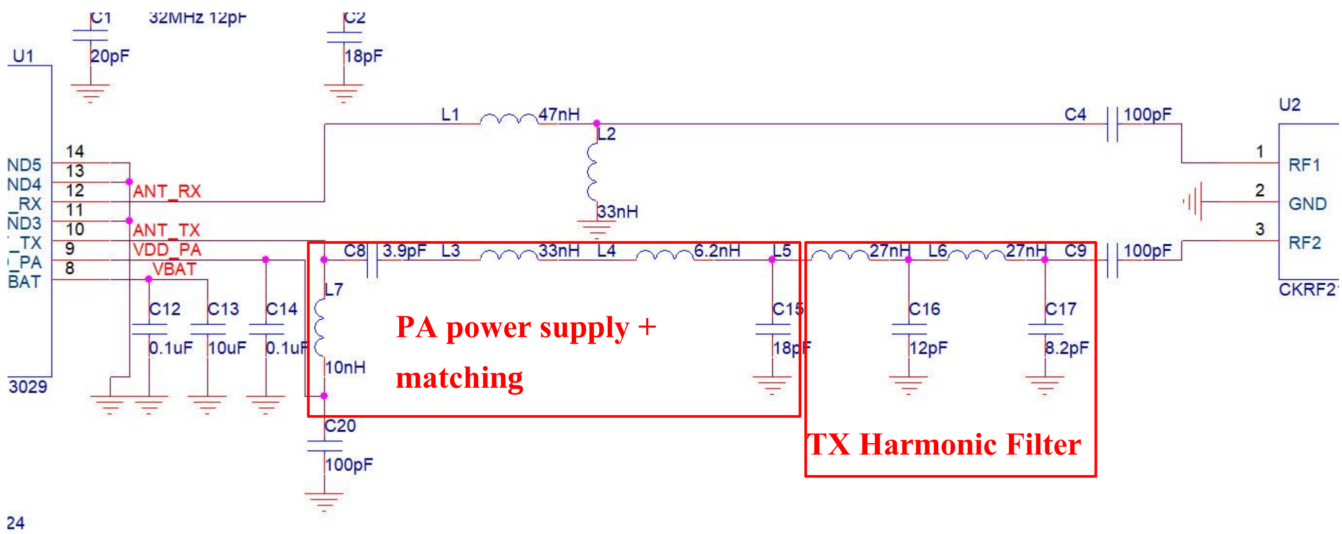
2. I2C and SPI multiplex pins, SCL multiplexes SPI's SCK, SDA multiplexes SPI's MOSI, CSN must be pulled high when using I2C, and the maximum rate of I2C is required to be lower than 15Mbps;

3. Interrupt IRQ, the default is low level output, and it will output high level when transmission and reception are successful.

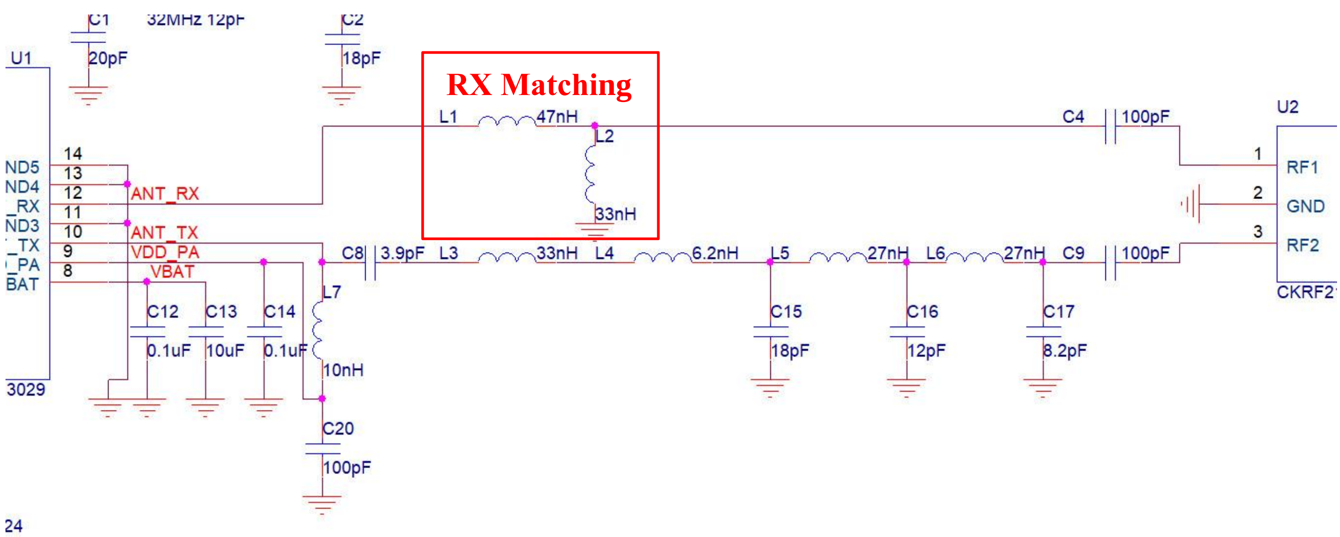
4. Please refer to the user manual for the GPIO function configuration. The key point is that GPIO11 can be reused as a CAD detection function pin, and GPIO10 can be configured as an external PA control pin.

1.5 RF front-end design requirements

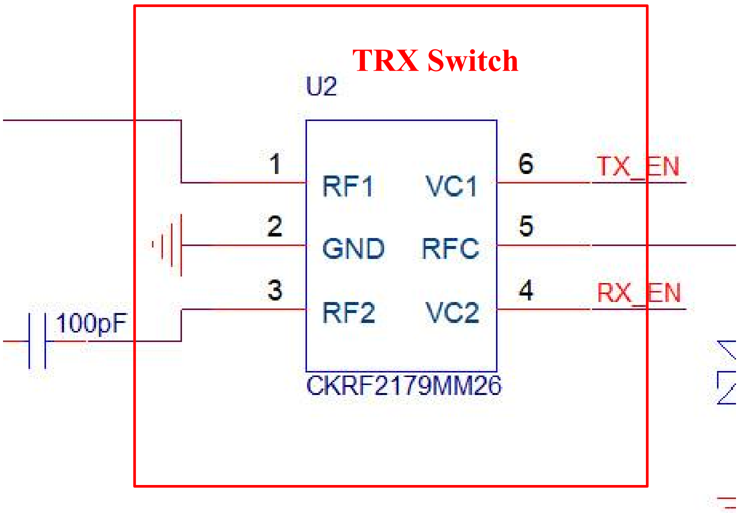
The chip TX-side circuit needs to complete the functions of powering the PA, matching, and filtering harmonics. PA powering, matching, and filtering harmonics correspond to the following circuits respectively.



The chip RX end circuit needs to complete the matching function, corresponding to the following circuit. Since DCDC will interfere with RX after operation, resulting in a slight loss of RX sensitivity, if there are high requirements for RF performance, it is recommended to use LDO mode.



There are two ways to combine the TX and RX of the chip. One is to combine through an external switch. In this way, the TX and RX performance can reach the best state; the other is to combine directly by adjusting the inductance and capacitance values. In this way, the TX and RX performance will be lost.
RX sensitivity will be grea



1.6 ESD design requirements

The ESD performance of the antenna end is worse than that of other pins, so an ESD protection diode needs to be reserved

1.7 Reference Schematic

1. Connect the middle EPAD to the outside ground through the GND pin on the top layer;

2. The DCDC ground needs to be connected to GND at a single point through a 0 resistor, and should not be directly connected to GND.

3. The inductor selection requires that the on-resistance is less than 100 m and the peak current is greater than 150mA (reference model: PIM252010-2R2MTS00).

4. L5, C16, L6, and C17 are for safety filter matching. If safety is not a concern, they can be removed.

5. The matching component values need to be fine-tuned according to the TR Switch and Layout;

6. DCDC mode is only recommended to be used in RX mode. Turning on DCDC in TX mode will affect safety regulations.

1.7.1 433MHz

1.7.1.1 Reference Schematic

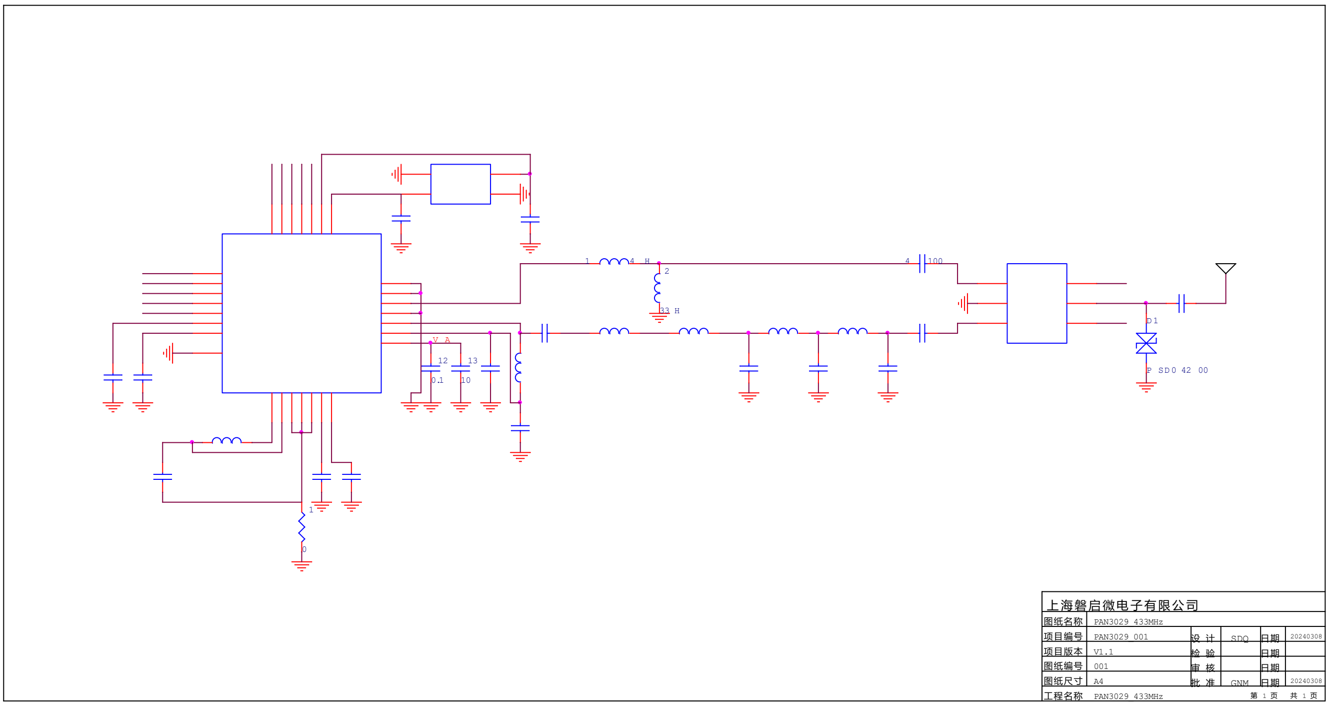


Figure 1-1 433MHz reference schematic

1.7.1.2 Reference BOM

Table 1- 2 LDO Mode 433MHz Reference BOM

Designator	Value	Description	Footprint
R1	0R		0402
C8	3.9pF	NPO, ±0.1pF, 50V	0402
C17	8.2pF	NPO, ±0.1pF, 50V	0402
C16	12pF	NPO, ±1%, 50V	0402
C2,C15	18pF	NPO, ±1%, 50V	0402
C1	20pF	NPO, ±1%, 50V	0402
C4,C7,C9,C20	100pF	NPO, ±5%, 50V	0402
C12,C14,C18,C19,C23,C24	0.1uF	X7R, ±10%, 16V	0402
C21	1uF	X7R, ±20%, 16V	0402
C13	10uF	X7R, ±20%, 16V	0603
D1	-	PESD0542U005	0402
L4	6.2nH	LQW15AN6N2G00D	0402
L7	10nH	LQW15AN10NG00D	0402

L5,L6	27nH	LQW15AN27NG00D	0402
L2,L3	33nH	LQW15AN33NG00D	0402
L1	47nH	LQW15AN77NG00D	0402
L8	2.2uH	PIM252010-2R2MTS00	2520
U1	PAN3029	-	QFN28_4*4
U2	CKRF2179MM26	RF switch	SOT-363
Y1	32MHz	Crystal, CL=12pF,±10PPM	SMD3225

1.7.2 433MHz_ Switchless

1.7.2.1 Reference Schematic

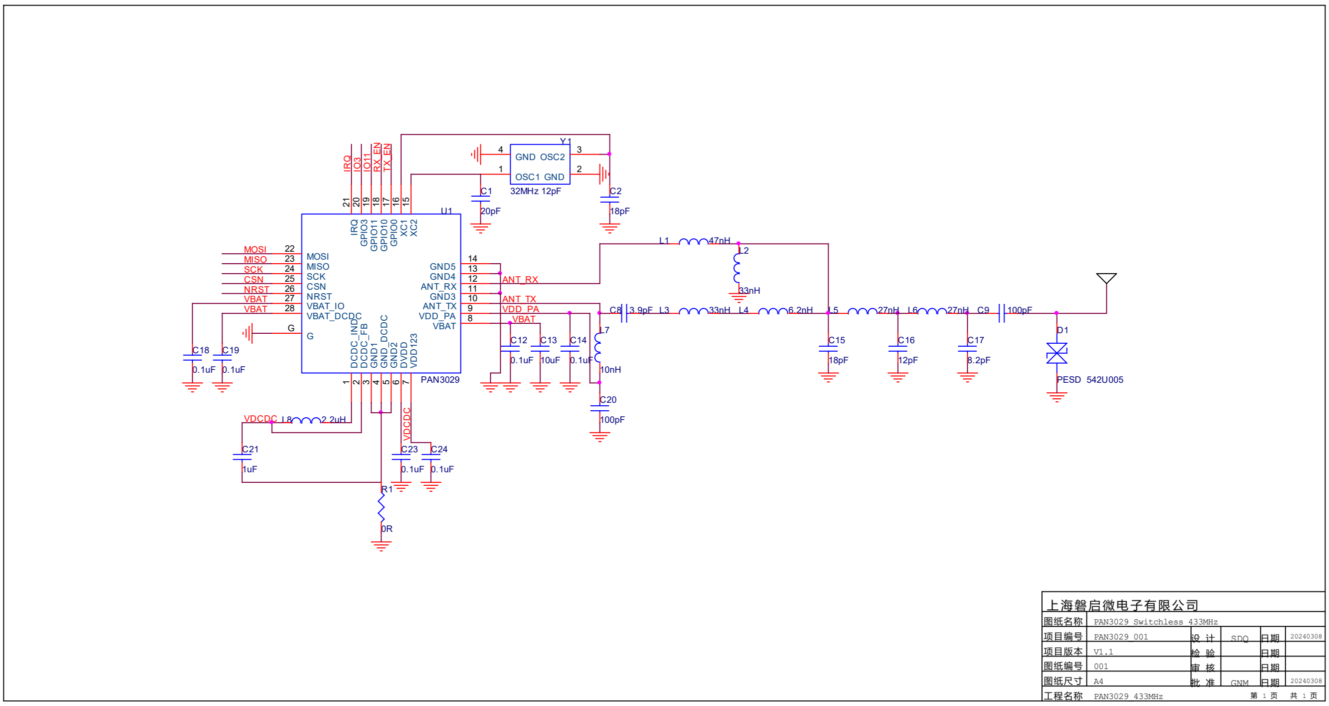


Figure 1-1 Switchless 433MHz reference schematic

1.7.2.2 Reference BOM

Table 1- 2 DCDC Mode 433MHz Reference BOM

Designator	Value	Description	Footprint
R1	0R	Chip resistor, ±1%	0402
C8	3.9pF	Chip capacitor, NPO, ±0.1pF,25V	0402
C17	8.2pF	Chip capacitor, NPO, ±0.1pF,25V	0402
C16	12pF	Chip capacitor, NPO, ±1%, 50V	0402
C2,C15	18pF	Chip capacitor, NPO, ±1%, 50V	0402
C1	20pF	Chip capacitor, NPO, ±1%, 50V	0402
C9,C20	100pF	Chip capacitor, NPO, ±5%, 50V	0402
C12,C14,C18,C19,C23,C24	0.1uF	Chip capacitor, X7R, ±10%, 16V	0402
C21	1uF	Chip capacitor, X7R, ±10%, 16V	0402
C13	10uF	Chip capacitor, X7R, ±20%, 16V	0603
D1	-	PESD0542U005	0402
L4	6.2nH	LQW15AN6N2G00D	0402
L7	10nH	LQW15AN10NG00D	0402

L5,L6	27nH	LQW15AN27NG00D	0402
L2,L3	33nH	LQW15AN33NG00D	0402
L1	47nH	LQW15AN47NG00D	0402
L8	2.2uH	PIM252010-2R2MTS00	2520
U1	PAN3029	-	QFN28_4*4
Y1	32MHz	Crystal, CL=12pF,±10PPM	SMD3225

1.7.3 490MHz

1.7.3.1 Reference Schematic

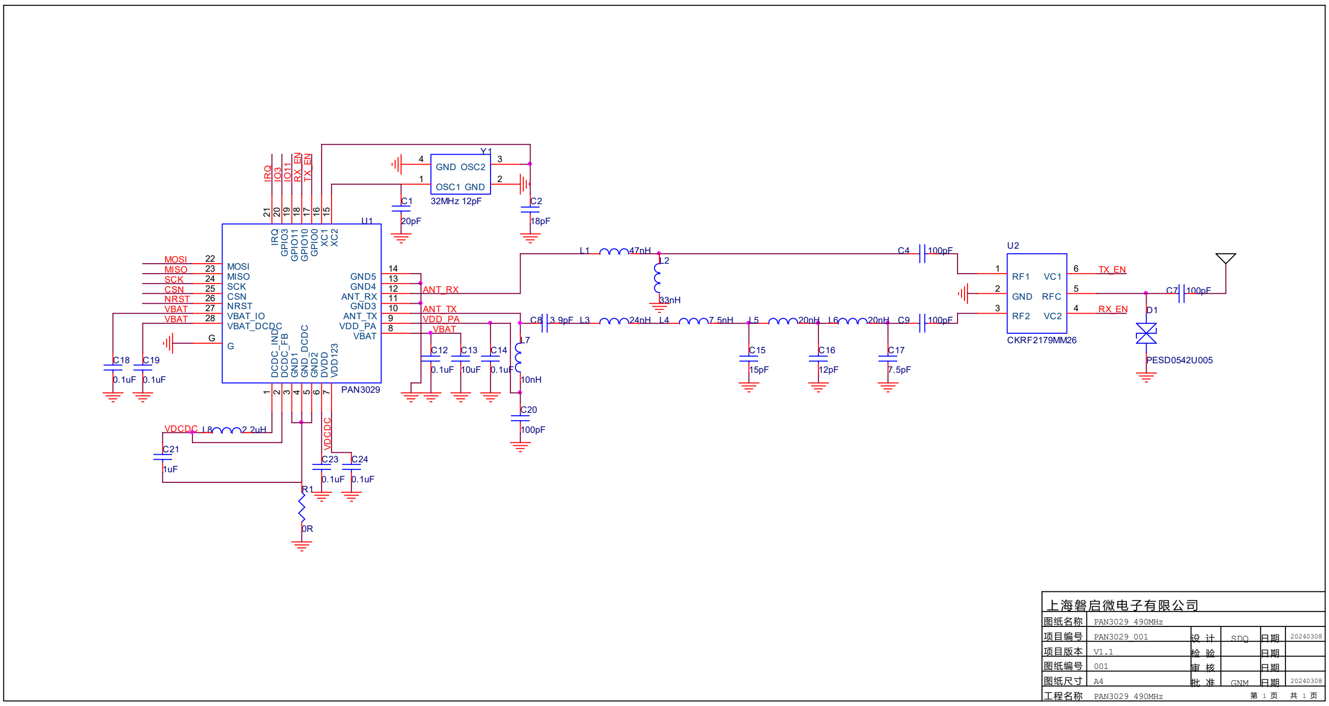


Figure 1-2 490MHz reference schematic

1.7.3.2 Reference BOM

表 1-3 490MHz 参考 BOM

位号	值	描述	封装
R1	0R	贴片电阻，±1%	0402
C8	3.9pF	贴片电容，NPO，±0.1pF，50V	0402
C17	7.5pF	贴片电容，NPO，±0.1pF，50V	0402
C16	12pF	贴片电容，NPO，±1%，50V	0402
C15	15pF	贴片电容，NPO，±1%，50V	0402
C2	18 pF	贴片电容，NPO，±1%，50V	0402
C1	20 pF	贴片电容，NPO，±1%，50V	0402
C4,C7,C9,C20	100pF	贴片电容，NPO，±5%，50V	0402
C12,C14,C18,C19,C23,C24	0.1uF	贴片电容，X7R，±10%，16V	0402
C21	1uF	贴片电容，X7R，±10%，16V	0402
C13	10uF	贴片电容，X7R，±20%，16V	0603
D1	-	PESD0542U005	0402
L4	7.5nH	LQW15AN7N5G00D	0402

L7	10nH	LQW15AN10NG00D	0402
L5,L6	20nH	LQW15AN20NG00D	0402
L3	24nH	LQW15AN24NG00D	0402
L2	33nH	LQW15AN33NG00D	0402
L1	47nH	LQW15AN47NG00D	0402
L8	2.2uH	PIM252010-2R2MTS00	2520
U1	PAN3029	-	QFN28_4*4
U2	CKRF2179MM26	射频开关	SOT-363
Y1	32MHz	贴片无源晶振，CL=12pF,±10PPM	SMD3225

1.7.4 490MHz_ Switchless

1.7.4.1 Reference Schematic

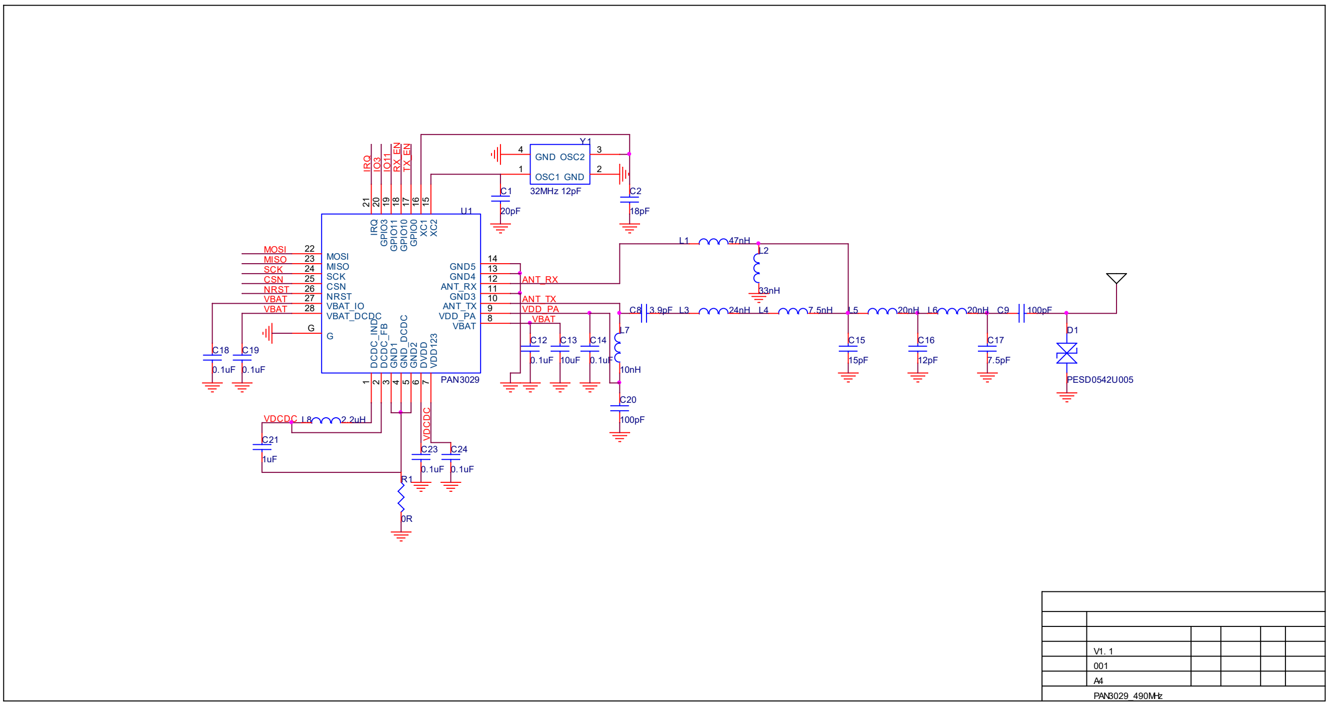


Figure 1-3 Switchless_490MHz reference schematic

1.7.4.2 Reference BOM

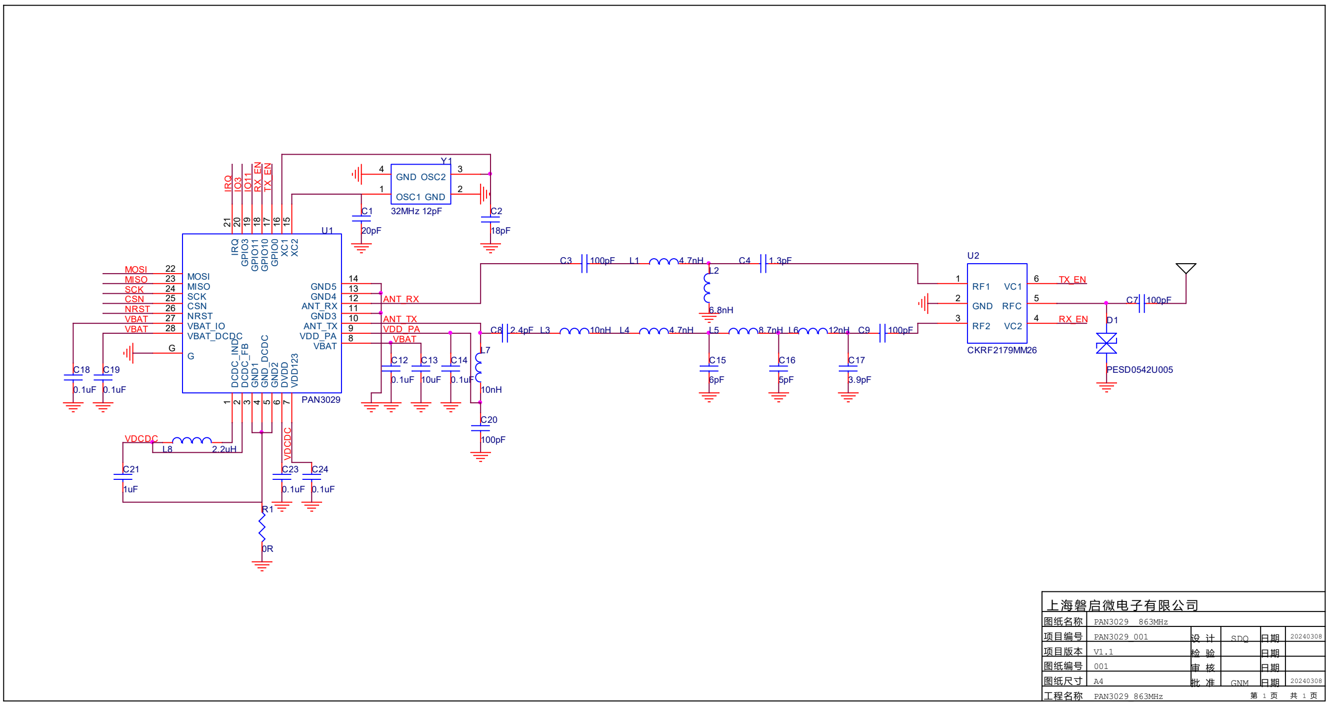
表 1-4 Switchless_490MHz 参考 BOM

位号	值	描述	封装
R1	0R	贴片电阻，±1%	0402
C8	3.9pF	贴片电容，NPO，±0.1pF，50V	0402
C17	7.5pF	贴片电容，NPO，±0.1pF，50V	0402
C16	12pF	贴片电容，NPO，±1%，50V	0402
C15	15pF	贴片电容，NPO，±1%，50V	0402
C2	18 pF	贴片电容，NPO，±1%，50V	0402
C1	20 pF	贴片电容，NPO，±1%，50V	0402
C9,C20	100pF	贴片电容，NPO，±5%，50V	0402
C12,C14,C18,C19,C23,C24	0.1uF	贴片电容，X7R，±10%，16V	0402
C21	1uF	贴片电容，X7R，±10%，16V	0402
C13	10uF	贴片电容，X7R，±20%，16V	0603
D1	-	PESD0542U005	0402
L4	7.5nH	LQW15AN7N5G00D	0402

L7	10nH	LQW15AN10NG00D	0402
L5,L6	20nH	LQW15AN20NG00D	0402
L3	24nH	LQW15AN24NG00D	0402
L2	33nH	LQW15AN33NG00D	0402
L1	47nH	LQW15AN47NG00D	0402
L8	2.2uH	PIM252010-2R2MTS00	2520
U1	PAN3029	-	QFN28_4*4
Y1	32MHz	贴片无源晶振，CL=12pF,±10PPM	SMD3225

1.7.5 863MHz

1.7.5.1 Reference Schematic



1.7.5.2 Reference BOM

表 1-5 863MHz 参考 BOM

位号	值	描述	封装
C4	1.3pF	贴片电容，NPO，±0.1pF，50V	0402
C8	2.4pF	贴片电容，NPO，±0.1pF，50V	0402
C17	3.9pF	贴片电容，NPO，±0.1pF，50V	0402
C16	5pF	贴片电容，NPO，±0.1pF，50V	0402
C15	6pF	贴片电容，NPO，±0.1pF，50V	0402
C2	18pF	贴片电容，NPO，±1%，50V	0402
C1	20pF	贴片电容，NPO，±1%，50V	0402
C3,C7,C9,C20	100pF	贴片电容，NPO，±5%，50V	0402
C12,C14,C18,C19,C23,C24	0.1uF	贴片电容，X7R，±10%，16V	0402
C21	1uF	贴片电容，X7R，±10%，16V	0402
C13	10uF	贴片电容，X7R，±20%，16V	0603
D1	-	PESD0542U005	0402
L1,L4	4.7nH	LQW15AN4N7G00D	0402

L2	6.8 nH	LQW15AN6N8G00D	0402
L5	8.7nH	LQW15AN8N7G00D	0402
L3,L7	10nH	LQW15AN10NG00D	0402
L6	12nH	LQW15AN12NG00D	0402
L8	2.2uH	PIM252010-2R2MTS00	2520
U1	PAN3029	-	QFN28_4*4
U2	CKRF2179MM26	射频开关	SOT-363
Y1	32MHz	贴片无源晶振,CL=12pF,±10PPM	SMD3225

1.7.6 915MHz

1.7.6.1 Reference Schematic

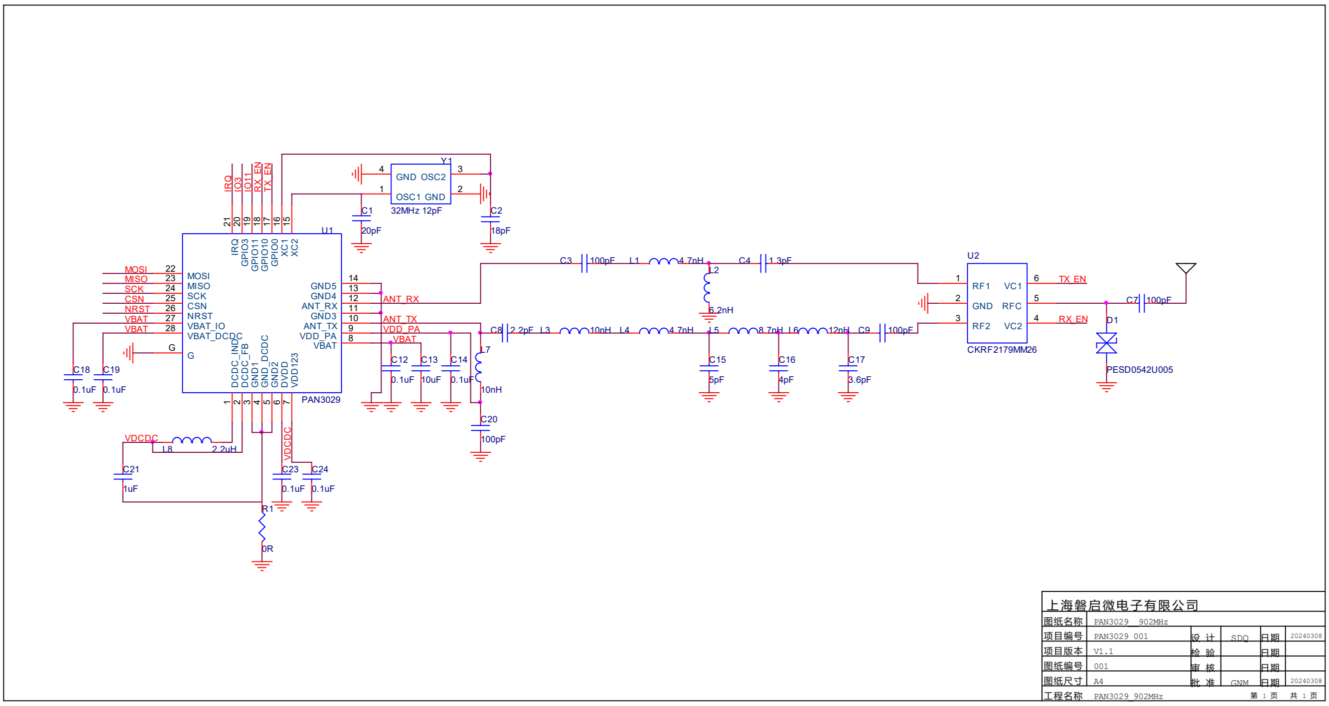


Figure 1-5 915MHz reference schematic

1.7.6.2 Reference BOM

表 1-6 915MHz 参考 BOM

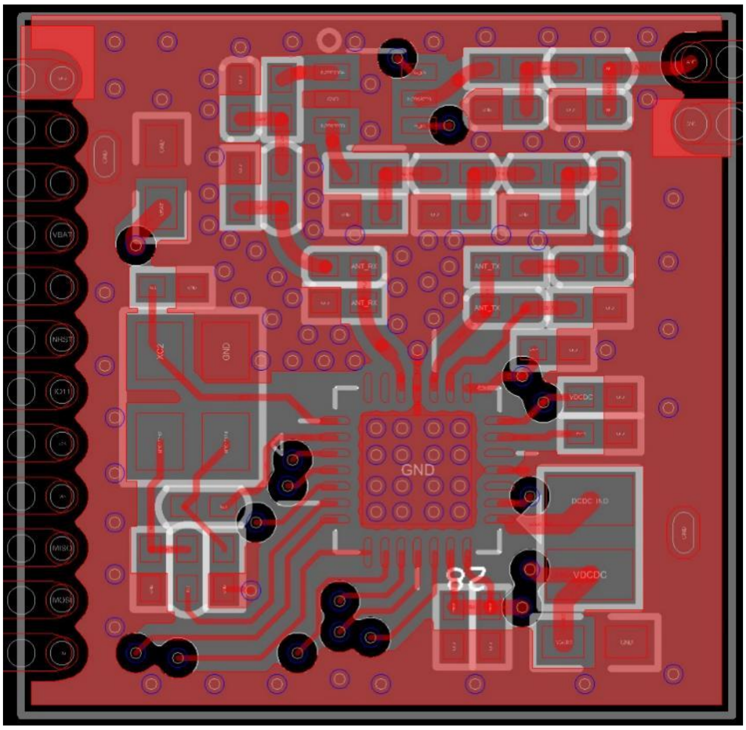
位号	值	描述	封装
C4	1.3pF	贴片电容，NPO，±0.1pF，50V	0402
C8	2.2pF	贴片电容，NPO，±0.1pF，50V	0402
C17	3.6pF	贴片电容，NPO，±0.1pF，50V	0402
C16	4pF	贴片电容，NPO，±0.1pF，50V	0402
C15	5pF	贴片电容，NPO，±0.1pF，50V	0402
C2	18pF	贴片电容，NPO，±1%，50V	0402
C1	20pF	贴片电容，NPO，±1%，50V	0402
C3,C7,C9,C20	100pF	贴片电容，NPO，±5%，50V	0402
C12,C14,C18,C19,C23,C24	0.1uF	贴片电容，X7R，±10%，16V	0402
C21	1uF	贴片电容，X7R，±10%，16V	0402
C13	10uF	贴片电容，X7R，±20%，16V	0603
D1	-	PESD0542U005	0402
L1,L4	4.7nH	LQW15AN4N7G00D	0402

L2	6.2nH	LQW15AN6N2G00D	0402
L5	8.7nH	LQW15AN8N7G00D	0402
L3,L7	10nH	LQW15AN10NG00D	0402
L6	12nH	LQW15AN12NG00D	0402
L8	2.2uH	PIM252010-2R2MTS00	2520
U1	PAN3029	-	QFN28_4*4
U2	CKRF2179MM26	射频开关	SOT-363
Y1	32MHz	贴片无源晶振，CL=12pF,±10PPM	SMD3225

2 PCB design requirements

2.1 Plate selection and special instructions

Regarding PCB design, it is recommended to use a four-layer board, which has better safety characteristics. If you need to consider factors such as cost, use a two-layer board layout. The middle EPAD needs to be connected to the outside ground through the top-layer GND pin, and the middle EPAD grounding vias should be as many as possible, such as 16.



2.2 DC/DC layout

1. The DCDC ground needs to be connected to GND at a single point through a 0 resistor, and should not be directly connected to GND.

2. The distance between the VSW pin and the inductor should be as short as possible and the wiring should be as thick as possible to improve the efficiency of DCDC.

2.3 Power and Ground Layout

The power line width is required to be above 0.5mm and withstand a transient current of 200 mA. Decoupling capacitors are placed close to the chip power pins, with small-value capacitors placed closer to the chip pins to better filter out high-frequency noise.

It is recommended that the power line and ground line be connected in a radial manner, with a single point for power/ground connection and separate routing. The power/ground routing of the RF chip should be separated from other chips or devices, and should be separately routed from the total reference power/ground line to prevent interference. If the power line is drawn from devices such as LDO or DCDC, it also needs to be routed separately and filtering measures should be taken. The GND pin at the bottom of the chip needs to be directly connected to the GND plane on the top layer of the circuit board.

In addition, it is recommended that the ground wire be connected to a ground wire with less noise or a total reference ground wire, and not to the ground wire or power wire of a strong signal or strong interference device, which can effectively reduce the operating noise of the entire printed circuit board.

2.4 Crystal Oscillator layout

1. To the extent that the area allows, keep a certain distance between the crystal and the chip and do a good job of heat insulation;

2. The pad of the direct-insertion crystal oscillator needs to ensure that the difference between the outer diameter and the inner diameter is more than 0.2mm;

3. To prevent the crystal oscillator signal from interfering with the RF signal, grounding treatment needs to be done on both sides of the crystal oscillator pad and the trace on the printed circuit board;

4. To prevent the crystal oscillator from being interfered by the transmission power of the antenna, a ground wire with a thickness of more than 0.5 mm should be used as a spacing strip between the antenna part on the printed circuit board and the crystal oscillator pad wiring part, and the crystal oscillator shell should be more than 3 mm away from the antenna.

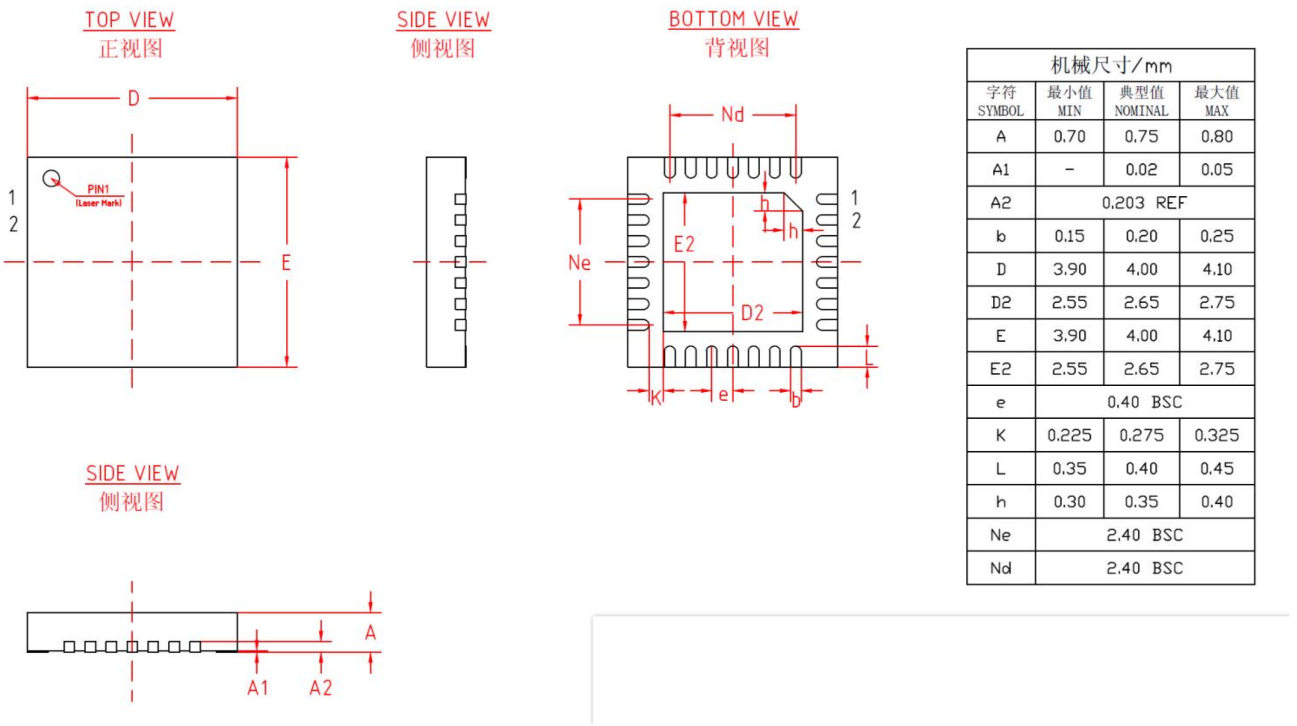
2.5 Control Lines layout

The SPI and IRQ lines of the control class need to reduce routing interference. The wiring should be short and have complete ground coverage on both sides of the wiring.

2.6 QFN package layout

PAN3029 (a chip in a 4mm×4mm QFN package) needs to have its pads underneath grounded. When the PCB is used as a library component, a large grounded pad needs to be added to the center of the chip. To ensure a good connection with the ground plane of the bottom layer of the double-layer PCB, 16 vias are recommended in the center of the pad. At the same time, the EPAD in the middle is connected to the outside ground through the NC pin on the top layer.

Avoid traces routing and components placing on the bottom layer of the PCB board under the PAN3029 chip, especially near the RF matching circuit. A complete ground plane can ensure good RF performance.



机械尺寸/mm			
字符 SYMBOL	最小值 MIN	典型值 NOMINAL	最大值 MAX
A	0.70	0.75	0.80
A1	-	0.02	0.05
A2	0.203 REF		
b	0.15	0.20	0.25
D	3.90	4.00	4.10
D2	2.55	2.65	2.75
E	3.90	4.00	4.10
E2	2.55	2.65	2.75
e	0.40 BSC		
K	0.225	0.275	0.325
L	0.35	0.40	0.45
h	0.30	0.35	0.40
Ne	2.40 BSC		
Nd	2.40 BSC		

Figure 2-1 PAN3029 QFN32 4*4 chip package dimensions

2.7 RF Matching Circuit layout

The RF matching structure should be designed strictly according to the recommended values in the schematic diagram. Due to layout differences, the component values can be fine-tuned.

1. In order to prevent the energy loss of the RF front end, the routing from the ANT pin to the antenna is short and is routed according to 50 ohm impedance (with a 0.3mm spacing from the surrounding ground, and a complete reference ground on the back). The ground around the RF matching part should be continuous and firm (with more holes), so that more transmission energy can be emitted from the antenna end;

2. In order to ensure the continuity of impedance, the back reference ground corresponding to the RF matching part should avoid placing components and wiring, and a complete ground plane is required;

3. It is recommended to use solid ground for covering;

4. The GND next to the antenna can be reserved for exposed copper to facilitate welding and debugging of the antenna;

5. The RF reference ground and EPAD need to be well connected;

6. It is prohibited to drill holes or change layers for RF lines.

3 Special Notes

3.1 PA output power

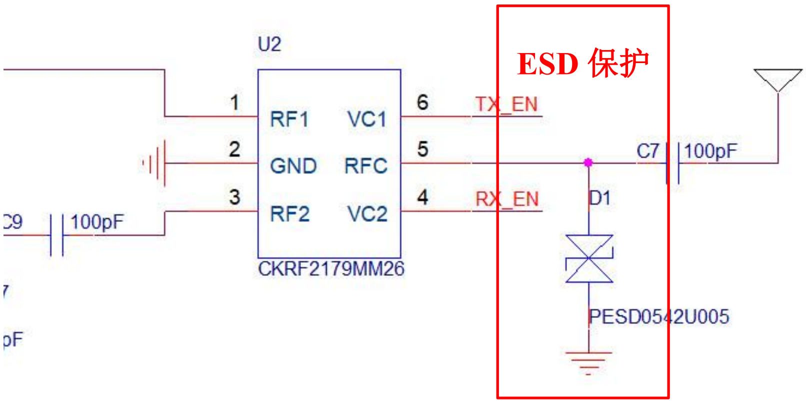
You need to strictly follow the configuration recommended by the SDK, otherwise it will introduce some risks and affect the application.

3.2 DC/DC receiving mode

Pins 3, 4, and 5 need to be connected to GND at a single point through a 0Ω resistor, otherwise the DC/DC will be turned on and affect the receiving sensitivity.

3.3 ESD protection

The ESD performance of the PAN3029 antenna pin is moderate. For industrial applications, it is recommended to add an ESD protection at the antenna port.



3.4 Special Layout Requirements

Regarding PCB design, it is recommended to use a four-layer board, which has better safety characteristics. If you need to consider factors such as cost, use a two-layer board layout. The middle EPAD needs to be connected to the outside ground through the GND pin on the top layer. At the same time, the middle EPAD grounding vias should be as many as possible, such as 16. The schematic diagram is shown in the figure below, otherwise there will be an abnormality of high power not being locked. A four-layer board layout is required when the output power of 863MHz and 915MHz exceeds 14dBm, otherwise there will be an abnormality of not being locked.